Static Structural Analysis of a Spur Gear Mechanism

Finite Element Analysis Report — SolidWorks CAD Model / ANSYS Workbench 2026 R1 (Static Structural)

1. Introduction and Objective

This report presents a static structural finite element analysis (FEA) of a two-gear spur gear mechanism consisting of a smaller driving gear (pinion) meshed with a larger driven gear. The gear pair was modeled as a solid assembly in SolidWorks and exported as a STEP file for analysis in ANSYS Workbench (Static Structural module). The objective of the study is to evaluate the structural response of the gear pair — specifically the total deformation, equivalent elastic strain, and equivalent (von-Mises) stress developed under the applied operating condition — and to identify the critical (highest-stressed) location in the assembly.

2. CAD Model

The gear mechanism was modeled as a two-gear spur gear pair in SolidWorks, consisting of a smaller pinion gear and a larger driven gear in mesh, sharing a common involute tooth profile and pressure angle so that conjugate action is maintained through the line of action. The solid model (including hub, bore, and full tooth geometry for both gears) was exported in STEP (AP214) format and imported directly into the ANSYS Workbench geometry cell for meshing and analysis. The detailed tooth and hub dimensions recorded on the SolidWorks drawing were used internally to size the mesh and verify the gear geometry prior to analysis and are not reproduced here.

3. Finite Element Model Setup

3.1 Analysis System

The analysis was carried out using the Static Structural solver in ANSYS Workbench 2026 R1 (Student license). The imported gear-pair geometry was meshed using the default quadrilateral-dominant surface/solid meshing scheme, with local mesh refinement automatically concentrated near the gear tooth contact (mesh) region, as seen in the finer, denser element pattern around the point of tooth engagement in the results plots below.

3.2 Boundary Conditions

Consistent with a standard meshing-gear static structural setup, the driven (larger) gear was constrained to represent its support condition, while a load/rotation was applied through the driving (smaller) gear so as to bring the tooth flanks of the two gears into contact at the mesh point. The solution was evaluated at a single time step (t = 1 s), representing the loaded, in-mesh condition of the gear pair.

3.3 Solution Outputs

Three results were requested from the Static Structural solution: Total Deformation, Equivalent Elastic Strain, and Equivalent (von-Mises) Stress, each evaluated over the full gear-pair assembly.

4. Results

4.1 Total Deformation

The total deformation contour (Figure 1) shows displacement concentrated almost entirely in the smaller (pinion) gear, increasing radially outward from its bore toward the tooth tips, while the larger gear — being the more rigidly supported member — shows negligible deformation. The maximum deformation of 6.1668×10^{-6} m occurs at the outer tooth tip of the pinion, farthest from its center of rotation, which is consistent with it behaving as the more flexible, overhung member of the pair.

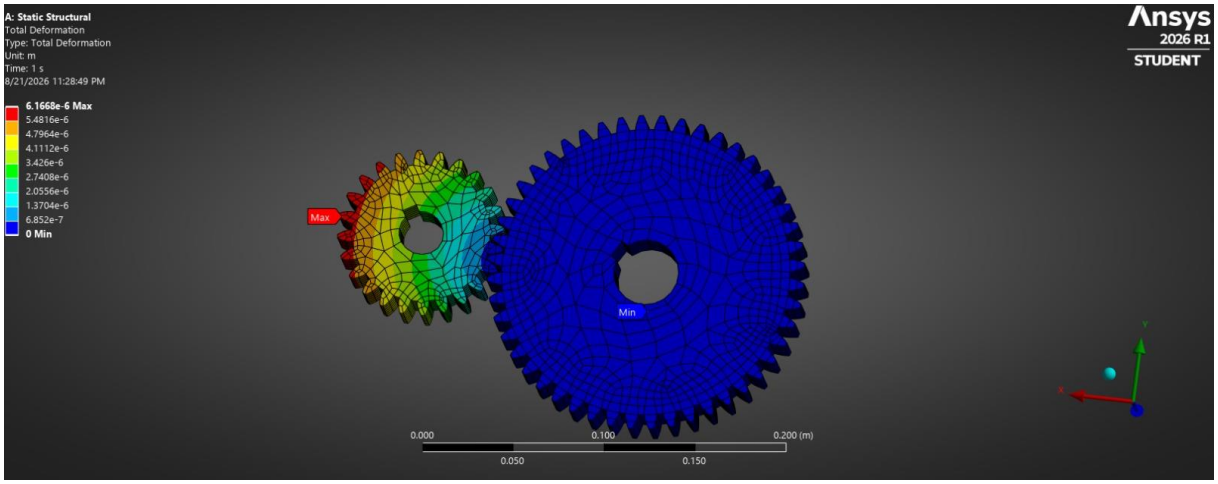


Figure 1 — Total deformation contour of the spur gear pair (max 6.1668 × 10^{-6} m at the pinion tooth tip).

4.2 Equivalent Elastic Strain

The equivalent elastic strain plot (Figure 2) shows strain sharply localized at the contact region between the meshing teeth of the two gears, with the peak value of 7.8558 × 10^{-5} m/m occurring precisely at the tooth flank contact (the "Max" marker). Away from this contact zone, strain levels fall rapidly to near zero throughout the rest of both gear bodies, indicating that the bulk of the material remains well within its elastic range and that the loading is highly localized, as expected for a Hertzian-type gear tooth contact.

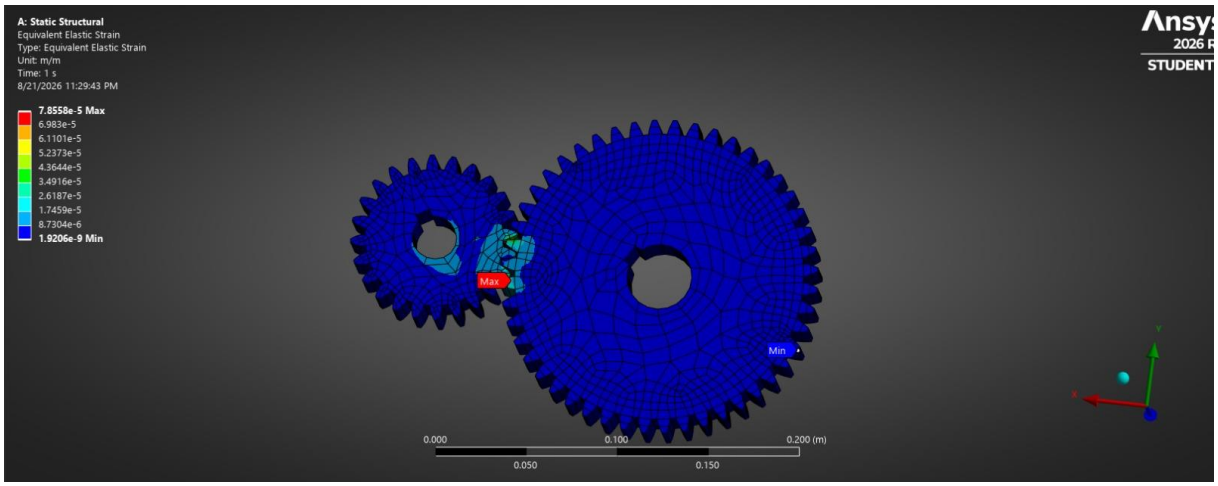


Figure 2 — Equivalent elastic strain contour, showing peak strain concentrated at the gear tooth mesh point (max 7.8558 × 10^{-5} m/m).

4.3 Equivalent (von-Mises) Stress

The equivalent (von-Mises) stress distribution (Figure 3) mirrors the strain pattern, with the maximum stress of 1.2379 × 10^{7} Pa (≈ 12.4 MPa) also occurring at the same tooth-contact location on the pinion gear. This is the expected result for a meshing gear pair, since the Hertzian contact stress at the line of tooth engagement is characteristically the highest-stressed location in a gear set, while the gear bodies away from the mesh point remain lightly loaded (stress close to the reported minimum).

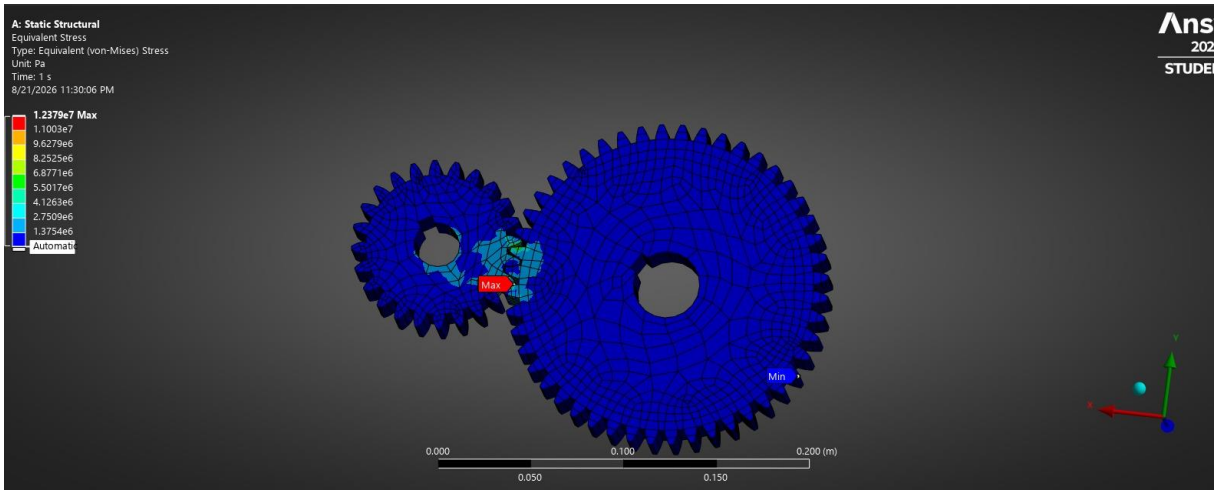


Figure 3 — Equivalent (von-Mises) stress contour of the meshing gear pair (max 1.2379 × 10^7 Pa at the tooth contact point).

4.4 Summary of Results

Result Quantity	Minimum	Maximum	Unit
Total Deformation	0	6.1668×10^{-6}	m
Equivalent Elastic Strain	1.9206×10^{-9}	7.8558×10^{-5}	m/m
Equivalent (von-Mises) Stress	≈ 0	1.2379×10^{7}	Pa

5. Discussion

All three field results — deformation, strain, and stress — consistently identify the tooth-contact (mesh) region of the pinion gear as the structurally critical location in the assembly. This is physically expected: the transmitted load passes through a small contact patch on the engaging tooth flanks, producing locally high Hertzian contact stresses and strains, while the remainder of each gear body, being far from the load path, experiences negligible structural response. The pinion (smaller gear) is more heavily affected than the larger gear in both deformation and stress terms, consistent with it carrying the same transmitted load over a smaller tooth cross-section and a more slender, overhung structure than the larger gear.

The peak von-Mises stress of about 12.4 MPa is well below typical yield strengths of common gear materials (e.g., structural/alloy steels, which typically yield in the 250–800 MPa range), suggesting that the gear pair, at the analyzed load level, operates with a considerable margin of safety against yielding. A more complete design check would additionally compare this stress against standard gear bending-fatigue (AGMA/Lewis) and surface-contact (Hertzian) allowable stress limits for the intended material and duty cycle.

6. Conclusion

- A two-gear spur gear mechanism was modeled in SolidWorks and analyzed under static structural loading in ANSYS Workbench 2026 R1.
- The maximum total deformation (6.17×10^{-6} m) occurs at the outer tooth tip of the smaller (pinion) gear.
- The maximum equivalent elastic strain (7.86×10^{-5} m/m) and maximum equivalent von-Mises stress (1.24×10^{7} Pa) both occur at the tooth-to-tooth contact point, consistent with expected gear-mesh contact behavior.
- The predicted stress level is well below typical allowable stresses for common gear materials, indicating an adequate structural margin under the analyzed loading condition.